Vital Sign Inference Using Contactless FMCW Radar

Areesh Sobhani [School of Electrical and Computer Engineering, Purdue University]

Mentors: Dr. Qiuye He, Dr. Yugyung Lee, Dr, Dianxiang Xu [Department of Computer Science and Information Technology, University of Missouri Kansas City]

INTRODUCTION

This project evaluates the feasibility and accuracy of mmWave-based biometric inference attacks across multiple subjects, highlighting their potential for medical applications as well as the need for privacy safeguards in future radar-based sensing platforms.

Medical Use Issues

In their extensive testing and commercial use, current methods of biometric data extraction generally rely on a 'point of contact'. While this is a sufficient way to test for biometric rates on the average person, there are many who are burdened extensive contact use, ranging from feelings of discomfort to direct injury.

Data Security Risks

In addition to healthcare applications, biometrics are increasingly being used in security and profiling applications. Considering this, the possibility of commercially accessible FMCW radars being used by a passive attacker to lift sensitive data relating to health, emotional state, or identity without consent.

BACKGROUND

Radar Basics

Frequency modulated continuous wave (FMCW) radars work on the principle of sending out and receiving a signal and using the difference between those two versions of the same signal to extract useful information. The transformed version of this intermediate frequency (IF) signal can indicate important position information about an object relative to the radar's viewpoint.

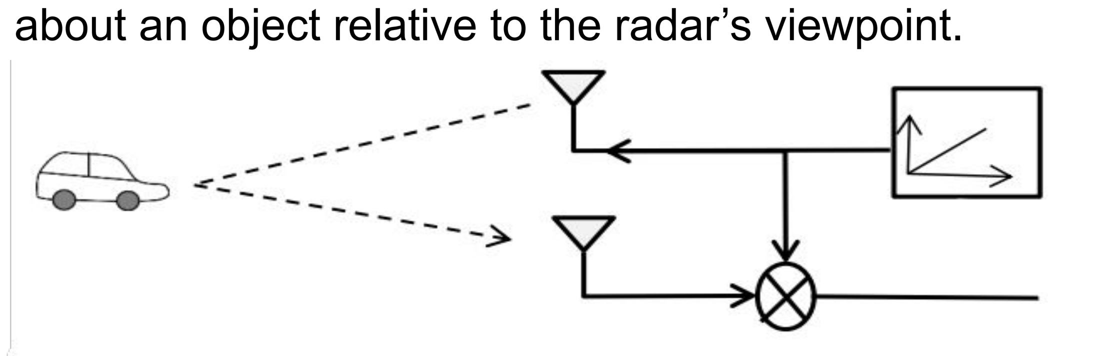

Figure 1. FMCW Radar Antenna System [1]

Biometric Rate Extraction

Micromovements like chest movement from breath is too small to be detected through range data alone, so the IF signal phase must be extracted through a second transform. Sending multiple 'chirp' signals out from the radar over time and processing them for phase relative to one another on reception allows the oscillation of the chest during breathing to be tracked.

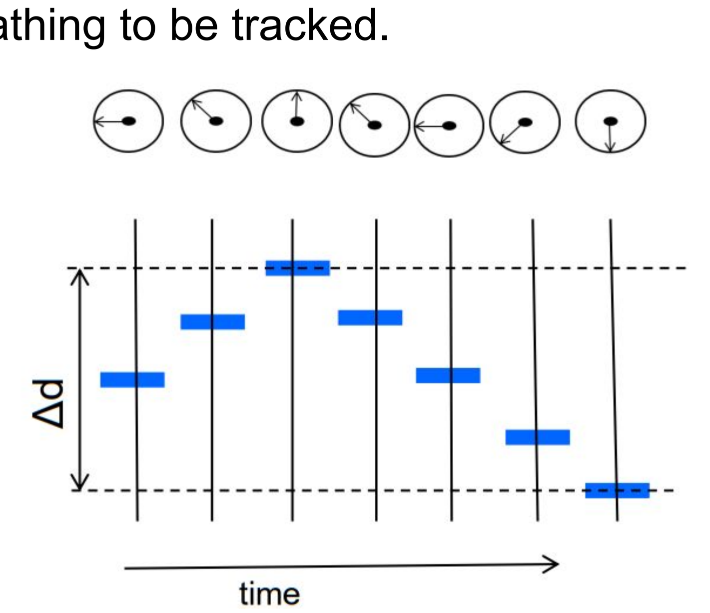

Figure 2. Tracked Signal Phase Across Time Domain [1]

METHODOLOGY

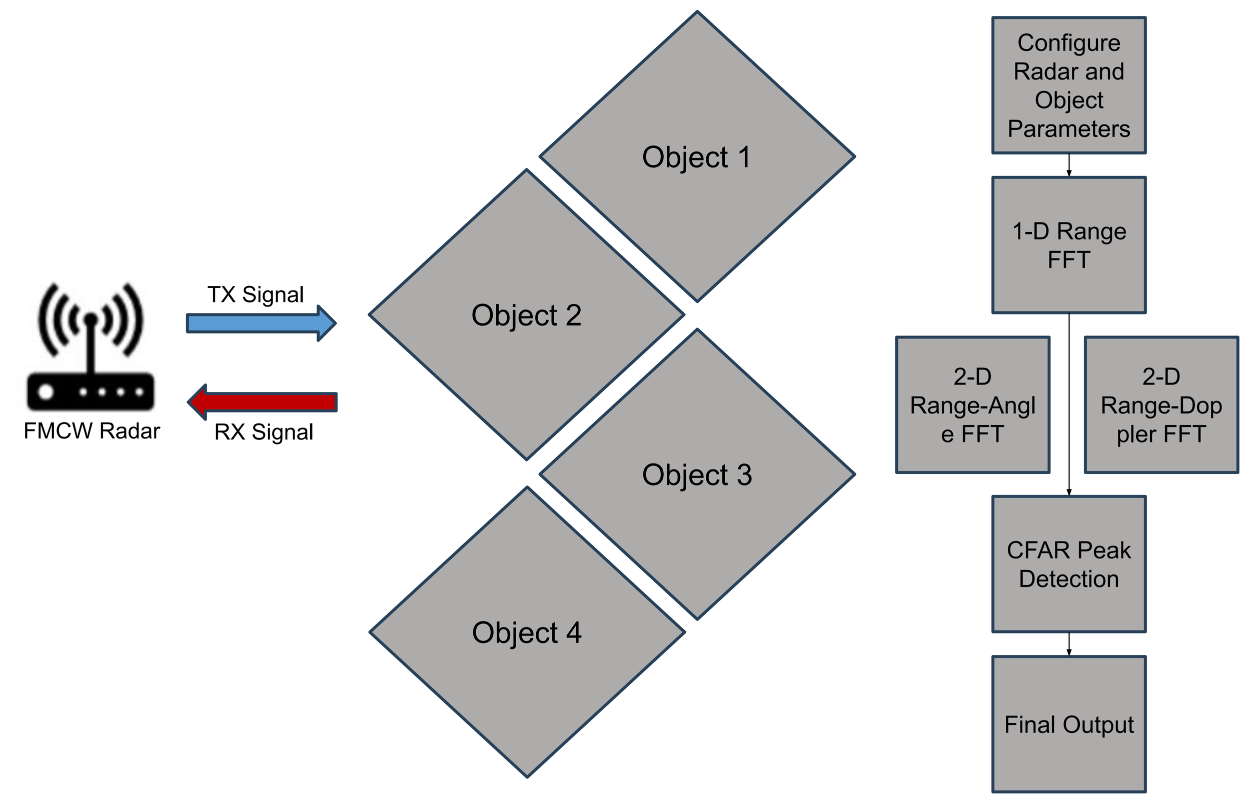

Figure 3. Ranging Simulation Setup and Pipeline

Configure Radar and Object Parameters:

- Set radar parameters: carrier frequency, bandwidth, chirp duration, sampling rate, and frame size.
- Set objects and parameters in view of simulated radar.
- Generate 'data cube' defined by chirp number per frame, sampling size, and number of Receive (RX) channels.

Data Extraction:

- Enhance signal-to-noise (SNR) ratio by coherent summation across all RX channels.
- Apply a series of fast Fourier transforms (FFTs) to resolve spatial and temporal dimensions.

Peak Detection and Final Output

- Constant False Alarm Rate (CFAR) algorithm creates varying 'threshold' relative to generated data to identify peaks.
- Data cube is indexed by peaks to output final data.

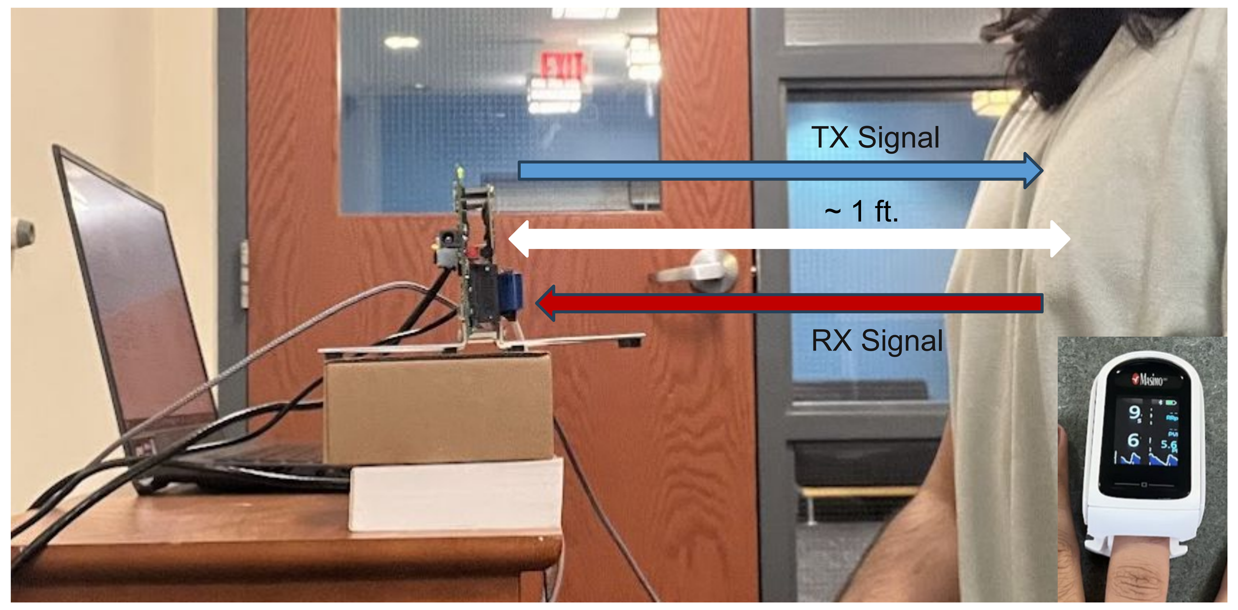

Figure 9. Biometrics Extraction Physical Setup

Physical Setup:

- Fix antenna array to chest-level and attach Masimo MightySat Rx for data validation.
- Configure radar parameters and data collection.

User Input, Data Reading and Reshaping:

- User is prompted for age and sex.
- Read signal data and reshape into 'data cube' format.

Data Extraction and Phase Processing:

- FFT resolves 'range' dimension.
- Extract complex component of signal in maximum range bin and unwrap to find signal phase across successive chirps to capture micromotion.

Denoising

- Independent Component Analysis (ICA) algorithm separates signal into statistically independent subcomponents.
- Frequencies component below minimum respiration rate and above maximum heart rate are filtered.

Signal Isolation, Validation, and Final Output

- Band-specific filters isolate respiration (0.2 – 0.33 Hz) and heart rate (0.83 – 2 Hz).
- Peaks in the defined frequency bands indicated vital signs, while peaks at the 1st harmonic of the heart rate enhance detection of the fundamental frequency.

SIMULATED OUTPUT

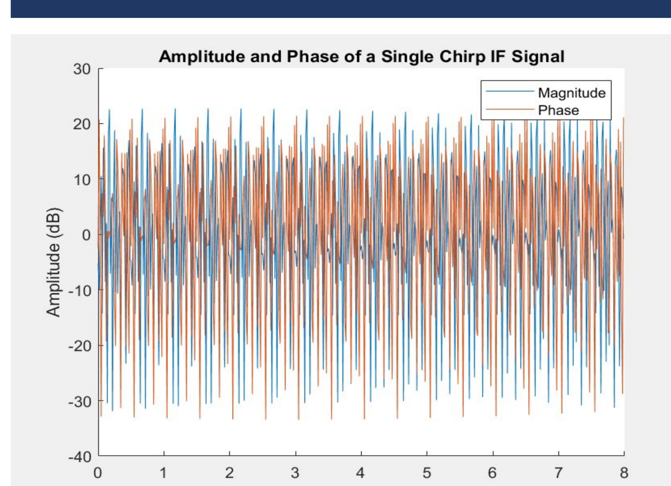

Figure 4. Simulated Radar Chirp

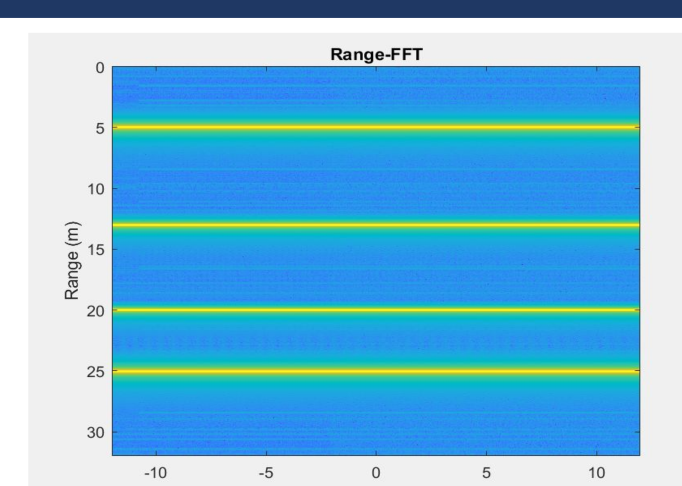

Figure 5. Simulated Range FFT

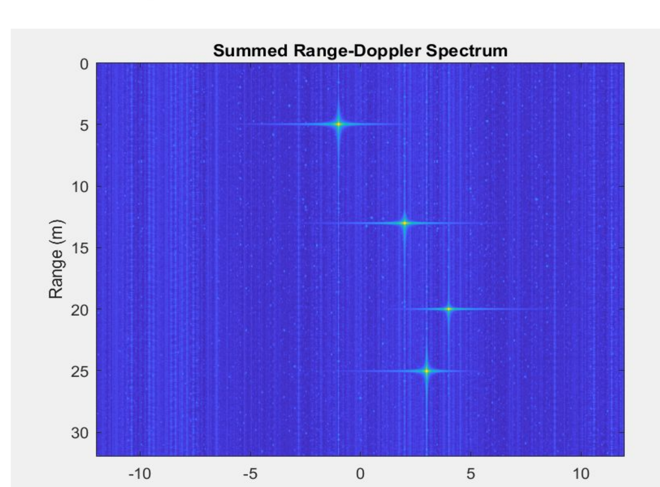

Figure 6. Simulated Range-Doppler Spectrum

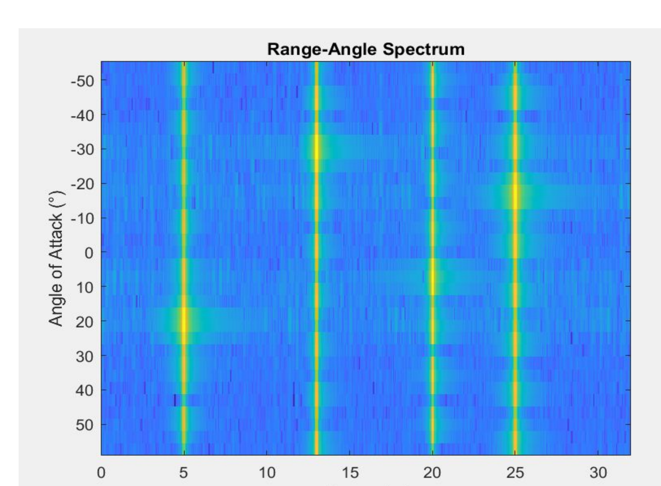

Figure 7. Simulated Range-Angle Spectrum

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There are 4 objects detected
Object 1 has a range of 5.00 m, a velocity of -1.01 m/s, and an angle of 21.49°.
Object 2 has a range of 13.07 m, a velocity of 1.99 m/s, and an angle of -28.65°.
Object 3 has a range of 20.07 m, a velocity of 4.00 m/s, and an angle of 7.16°.
Object 4 has a range of 25.02 m, a velocity of 3.00 m/s, and an angle of -17.90°.
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Figure 8. Simulated CFAR Text Output

EXPERIMENTAL RESULTS AND CASE STUDIES

Table 1. Average Percent Error for Extraction Rates Across Different Builds

	Average % Error of Heart Rate Extraction	Average % Error of Respiratory Rate Extraction
Subject 1 (F, 5'0", 146 lbs.)	11.79	26.91
Subject 2 (M, 6'2", 213 lbs.)	13.66	29.58
Subject 3 (M, 5'9", 135 lbs.)	14.84	74.08
Subject 4 (M, 5'11, 192 lbs.)	10.11	13.80

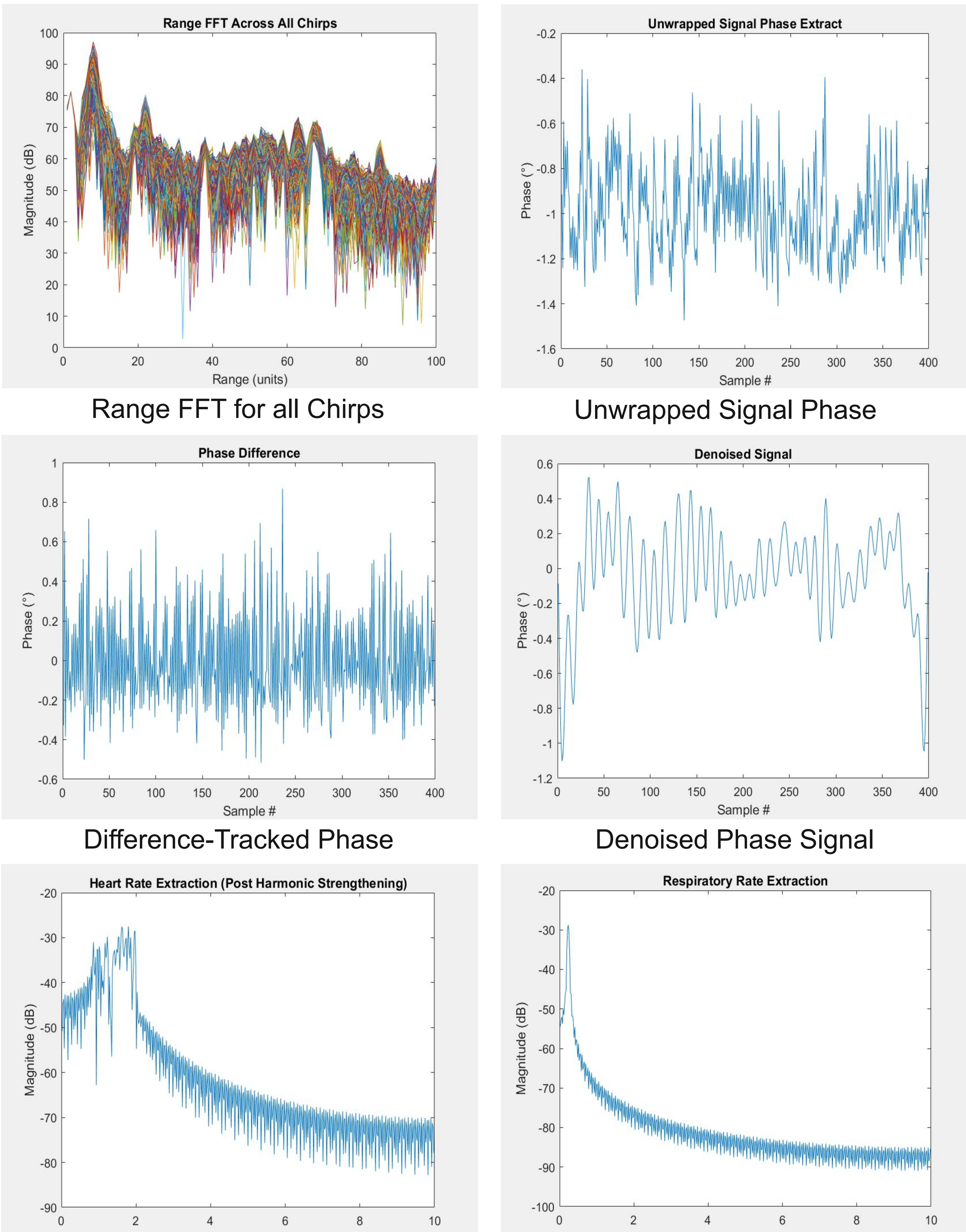
Table 2. Case Study of Closest Predictions – Pulse Extraction

	Predicted Heart Rate (BPM)	Actual Heart Rate (BPM)	% Error
Subject 1, Trial #5	94.92	94.25	0.71
Subject 2, Trial #3	69.14	69.50	0.52
Subject 3, Trial #1	69.14	69.55	0.59
Subject 4, Trial #2	55.08	55.25	0.31

Table 3. Case Study of Closest Predictions – Respiratory Rate Extraction

	Predicted Respiration (Breaths Per Minute)	Actual Respiration (Breaths Per Minute)	% Error
Subject 1, Trial #5	14.06	17.35	18.96
Subject 2, Trial #5	16.41	19.00	13.63
Subject 3, Trial #1	15.23	10.20	49.31
Subject 4, Trial #1	14.06	15.00	6.27

SIGNAL PIPELINE



Range FFT for all Chirps
Unwrapped Signal Phase
Difference-Tracked Phase
Denoised Phase Signal
Pulse Extraction
Respiration Extraction

Figure 10. Chest-Tracked Motion Data Pipeline

CONCLUSIONS

Key Takeaways:

- Pulse extraction across body types and sex is generally consistent, with absolute error ranging from around 10-15%.
- Respiration extraction varies significantly, with the most extreme absolute error sitting around 75%, most likely due to incorrect over-filtering of the signal.

Future Work:

- Defense methods against unsolicited biometrics extraction.
- User interface with continuous tracking and optimized filtering for increased accuracy and easier use.
- Data encryption and compression.

REFERENCES

[1] Rao, S. (n.d.). *Introduction to mmwave Sensing: FMCW Radars*. Texas Instruments. https://www.ti.com/content/dam/videos/external-videos/en-us/2/3816841626001/5415203482001.mp4/subassets/mmwaveSensing-FMCW-offlineviewing_0.pdf

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